Experiment 4

OLAP Operations (Slice, Dice, Drill-up & Drill-down) using Microsoft Power BI

AIM:

To perform Online Analytical Processing (OLAP) operations using Microsoft Power BI by creating multidimensional reports and applying Slice, Dice, Drill-down, and Drill-up operations on the Global Superstore dataset for analytical decision-making.

OBJECTIVES:

- To understand the basic concepts of OLAP in Business Intelligence.
- To perform Slice and Dice operations on business data.
- To perform Drill-down and Drill-up operations for detailed analysis.
- To analyze sales and profit across different dimensions.
- To create interactive reports using Power BI.
- To derive useful business insights from multidimensional data.

TOOLS / SOFTWARE REQUIRED:

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor – built into Power BI Desktop
- DAX (Data Analysis Expressions)

DATASET USED: Global Superstore Dataset.xlsx.

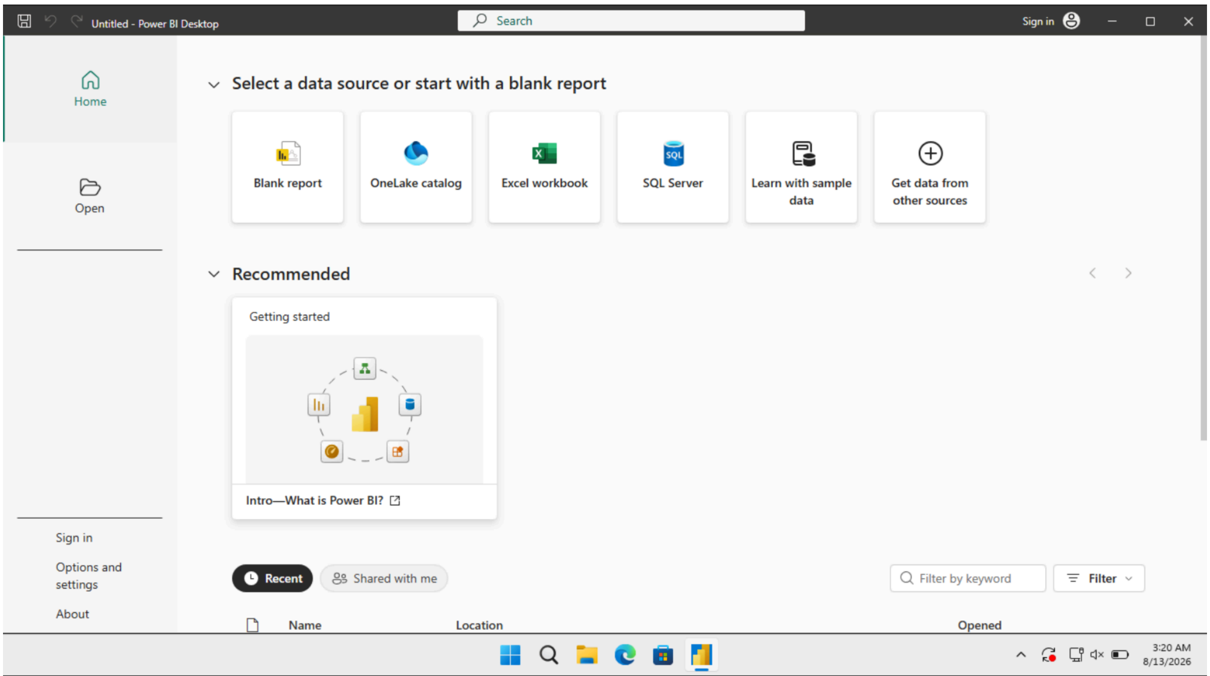
THEORY:

OLAP (Online Analytical Processing) is a technique used in Business Intelligence to analyze large amounts of data from different perspectives. It helps users explore business information and identify trends, patterns, and relationships for decision making. Multidimensional Analysis allows business data to be viewed using different dimensions such as Time, Region, Category, Product, and Customer. Measures such as Sales, Profit, and Quantity can be analyzed across these dimensions. Slice selects a particular value from one dimension to view a specific part of the dataset. For example, analyzing sales only for a particular year. Dice selects multiple values from different dimensions to create a smaller subset of data. For example, analyzing sales for selected regions and product categories.

PROCEDURE:

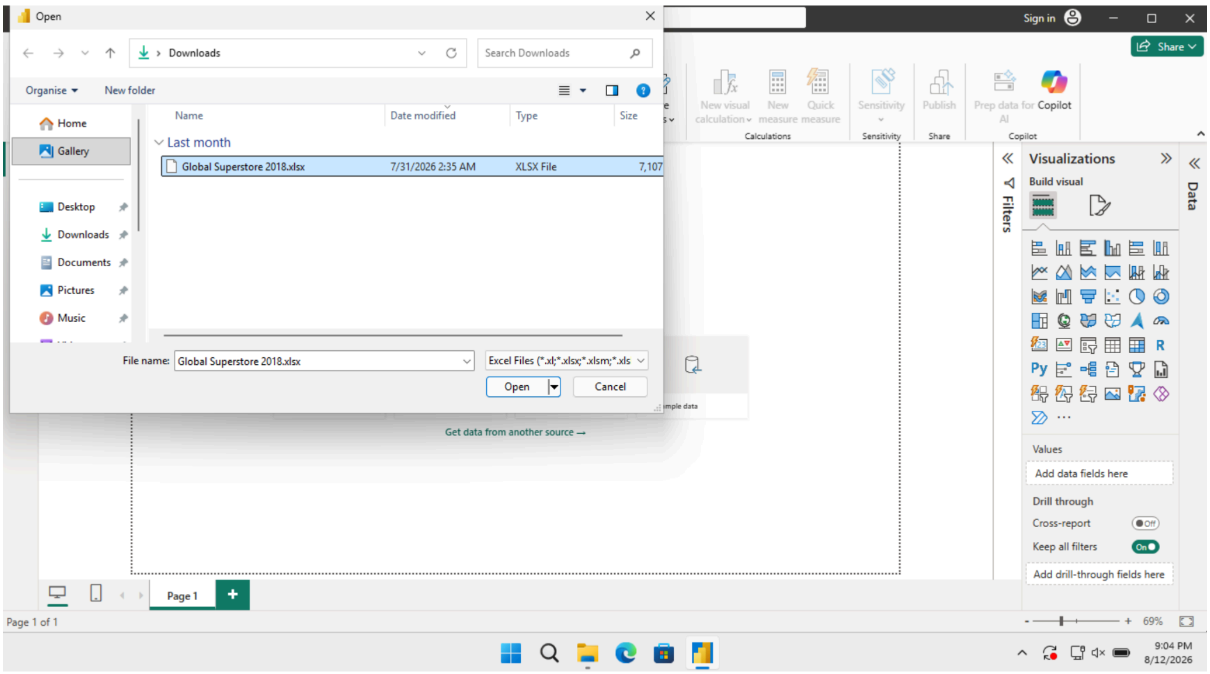
Part A: Load Dataset

Step 1: Open Microsoft Power BI Desktop.



Step 2: Import the Global Superstore dataset.

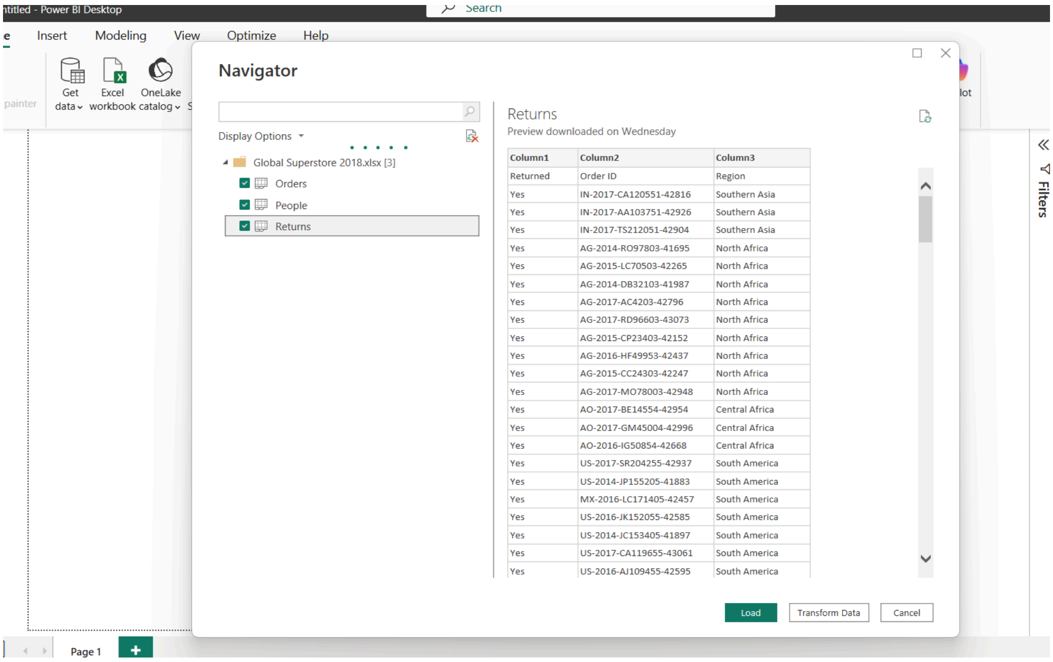
Home > Get Data > Excel Workbook Select:



Click Load.

Step 3: Select the required worksheets.

Import the following worksheets:

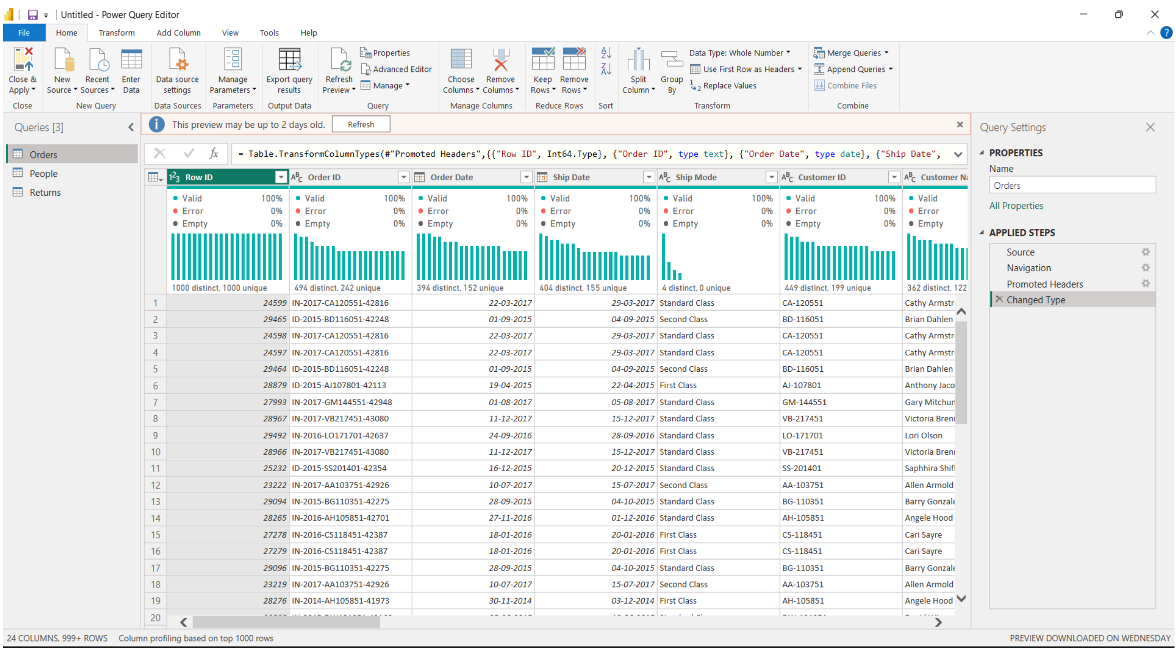


• Orders
• Returns
• People
Verify that the imported tables appear in Power BI.

Part B: Identify Fact and Dimension Tables
Step 4: Analyse the Orders table.
Examine the Orders table and identify measurable business attributes such as:

- Sales
- Profit
- Quantity
- Discount

These attributes represent measurable business events and are suitable for the Fact Table.
Step 5: Identify descriptive attributes.



Identify descriptive attributes related to:

- Customer
- Product
- Region
- Date

These attributes are used to create Dimension Tables.

Step 6: Categorize the data. Organize the attributes into:

A. Fact Table: Fact_Sales

B. Dimension Tables: Dim_Customer, Dim_Product, Dim_Region, Dim_Date

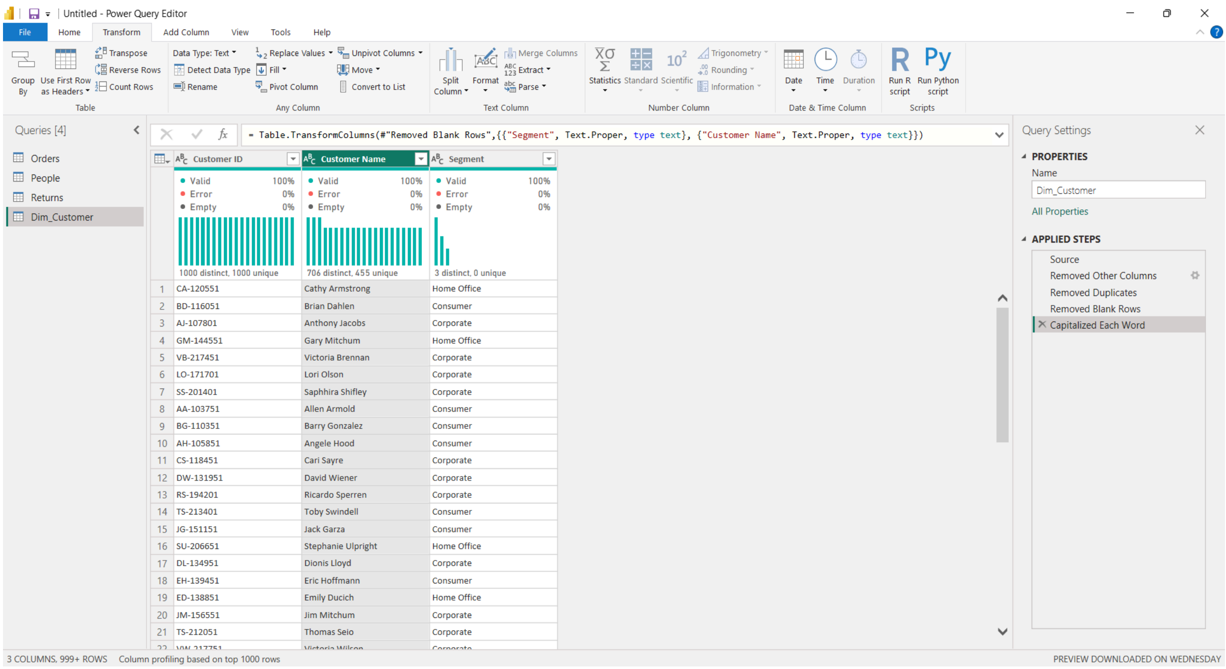
This classification forms the basis of the Star Schema.

Part C: Create Dimension Tables

Step 7: Open Power Query Editor.

Home > Transform Data

Step 8: Create Dim_Customer.

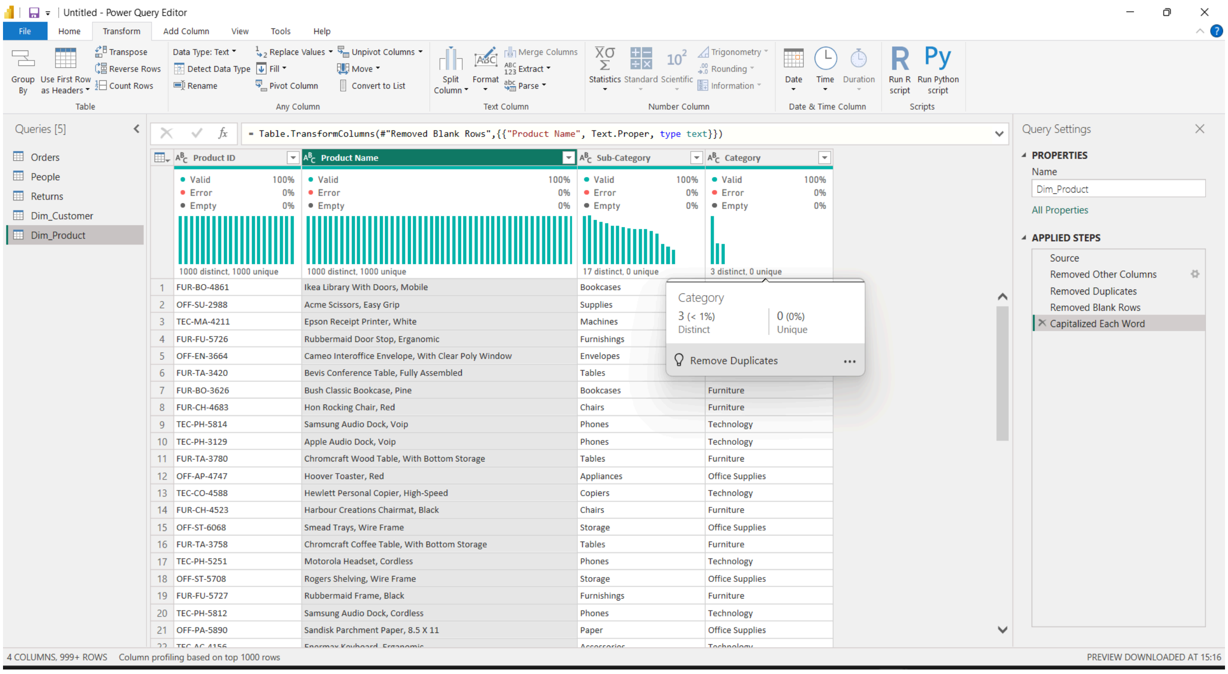


Create a customer dimension containing:

- Customer ID
- Customer Name
- Segment

Remove duplicate records from the dimension.

Step 9: Create Dim_Product.



Create a product dimension containing:

- Product ID
- Product Name
- Category
- Sub-Category

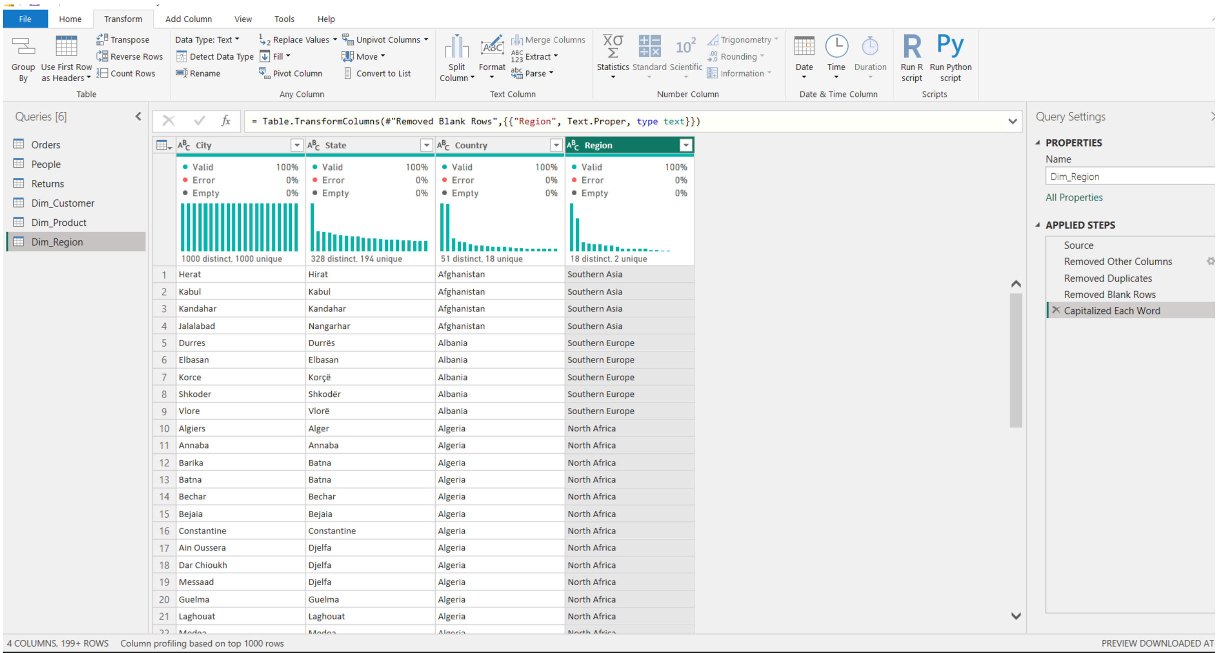
Remove duplicate records.

Step 10: Create Dim_Region.

Create a region dimension containing:

- Region
- Country
- State
- City

Remove duplicate records.

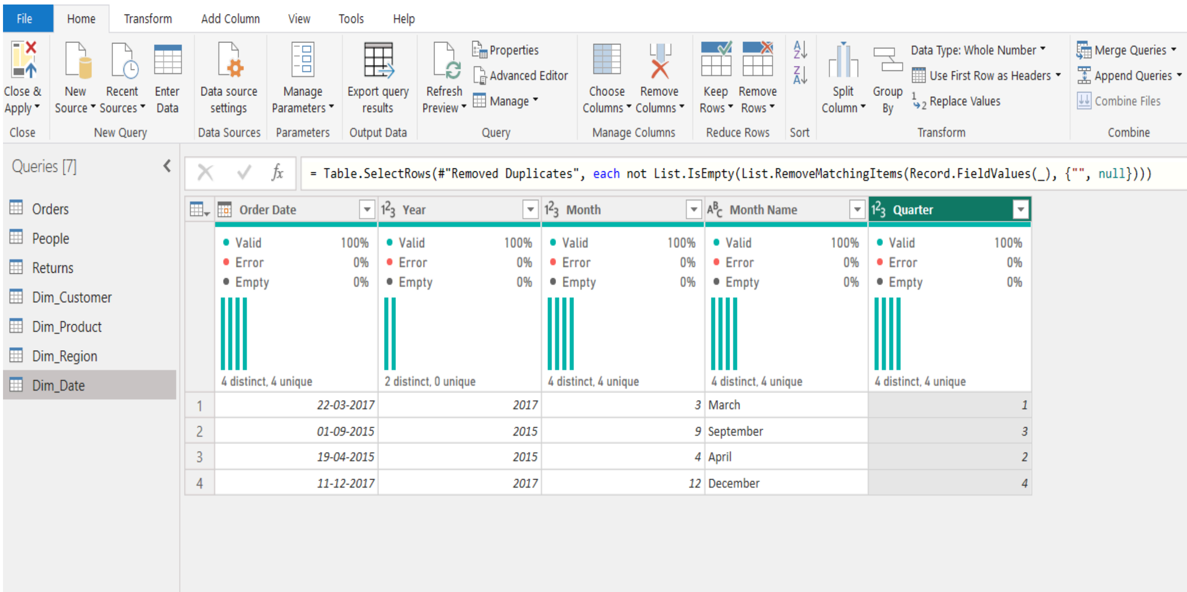


Step 11: Create Dim_Date.

Create a date dimension containing:

- Order Date
- Year
- Quarter
- Month

Ensure that the date-related fields are correctly formatted.



Step 12: Remove duplicate records.

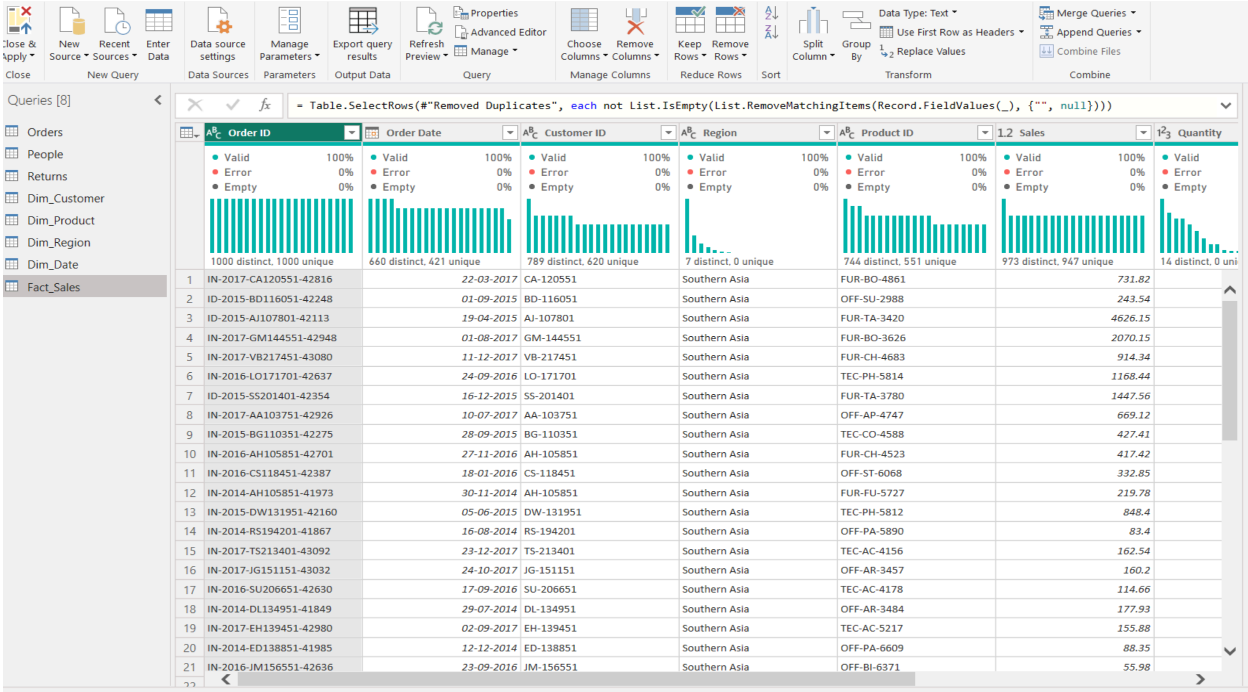
Remove duplicate records from all dimension tables and verify that the key fields contain unique values.

Part D: Create Fact Table

Step 13: Create Fact_Sales.

Create the central Fact Table containing:

- Order ID
- Customer ID
- Product ID
- Region
- Order Date
- Sales
- Quantity
- Profit
- Discount



Step 14: Verify key columns.

Verify that all key columns required to establish relationships with the Dimension Tables are present and correctly formatted.

The resulting Fact_Sales table acts as the central table of the Star Schema.

Part E: Build Star Schema

Step 15: Open Model View.

Select the Model View icon from the left navigation pane in Power BI.

Step 16: Create relationships.

Create relationships between:

• Fact_Sales ↔ Dim_Customer

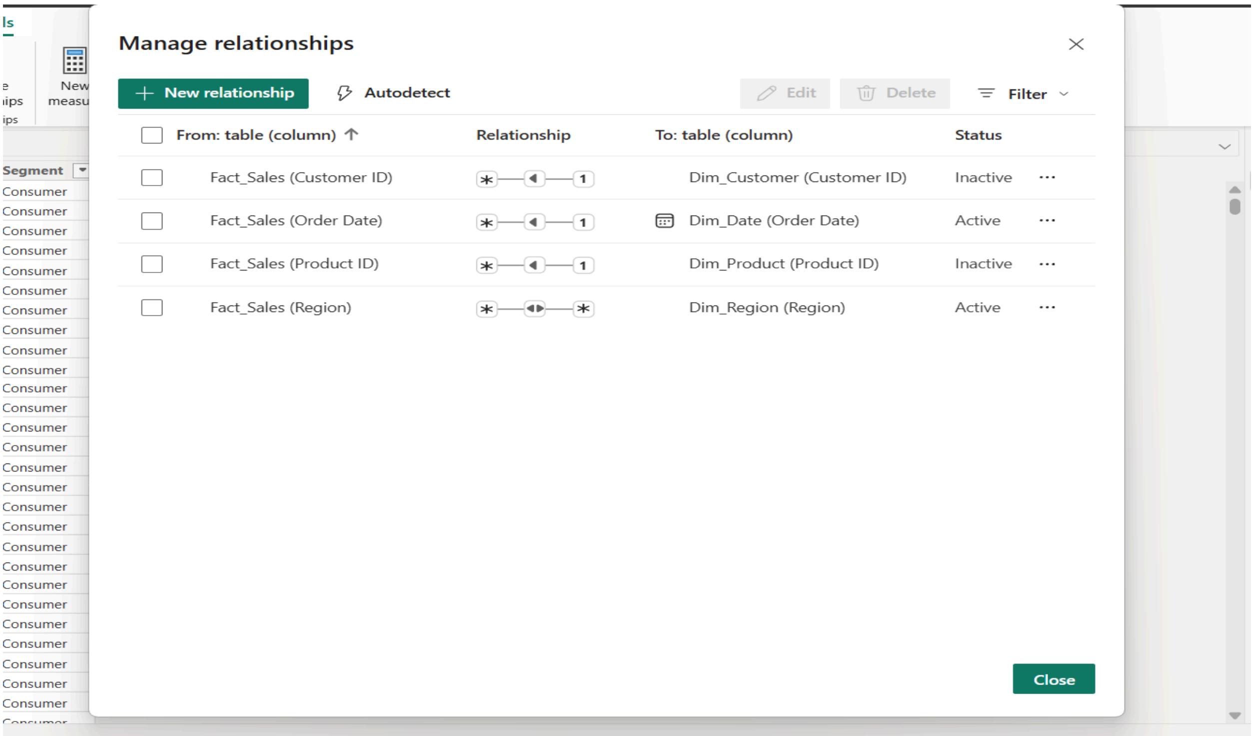
• Fact_Sales ↔ Dim_Product

• Fact_Sales ↔ Dim_Region

• Fact_Sales ↔ Dim_Date

Configure relationships.

Set the relationships appropriately as Many-to-One, with the Fact Table on the many side and the corresponding Dimension Table on the one side.



Step 18: Arrange the model.

Arrange the tables visually so that:

Dim_Region

|

▼

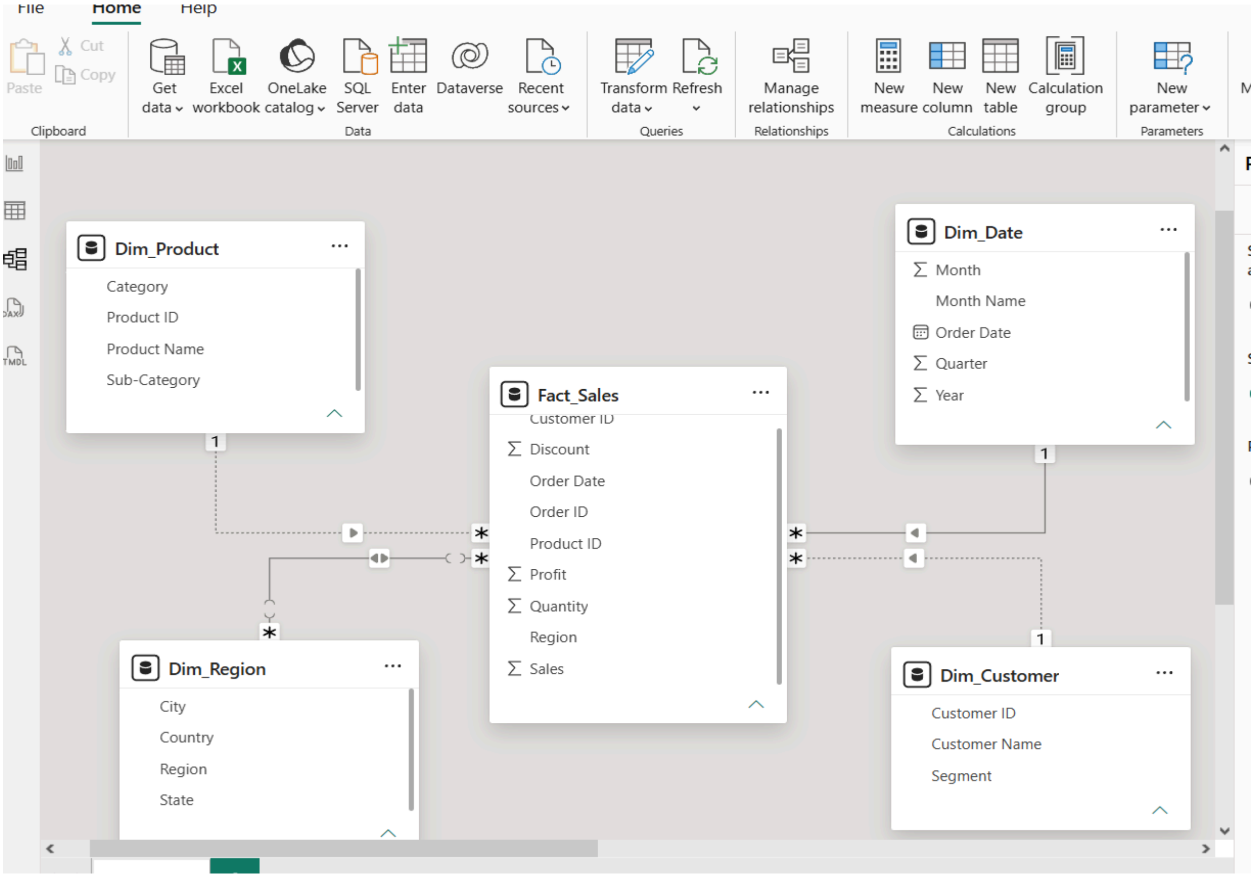
Dim_Customer ——► **Fact_Sales** ◄—— **Dim_Product**

▲

|

Dim_Date

This forms the Star Schema with Fact_Sales at the centre and the Dimension Tables . surrounding it.



Part F: Validate Model and Create DAX Measures

Step 19: Create Power BI visualizations.

Create visualizations using the dimensional model to analyse the data.

Examples include:

- Sales by Category
- Profit by Region
- Sales Trend by Year

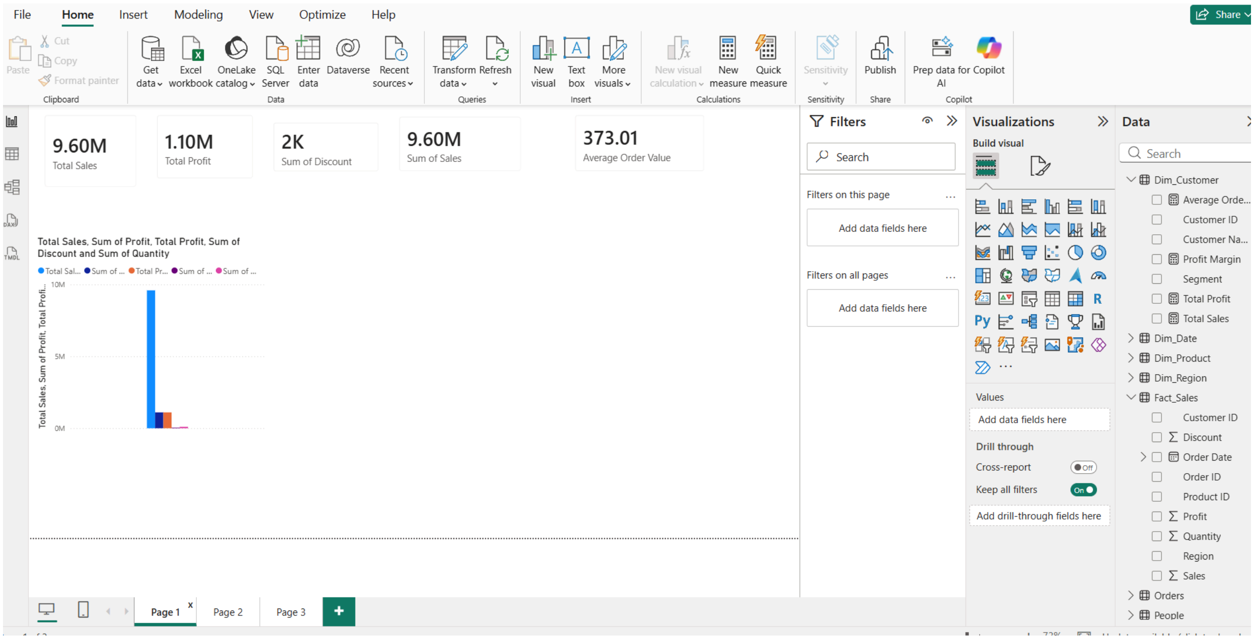
The reference specifies these as examples of visualizations for validating the model.

Step 20: Create DAX measures.

Create the following measures:

1. Total Sales = SUM(Fact_Sales[Sales])
2. Total Profit = SUM(Fact_Sales[Profit])
3. Profit Margin = DIVIDE(SUM(Fact_Sales[Profit]),SUM(Fact_Sales[Sales]))
4. Average Order Value = DIVIDE([Total Sales],DISTINCTCOUNT(Fact_Sales[Order ID]))

Step 21: Validate the Star Schema.

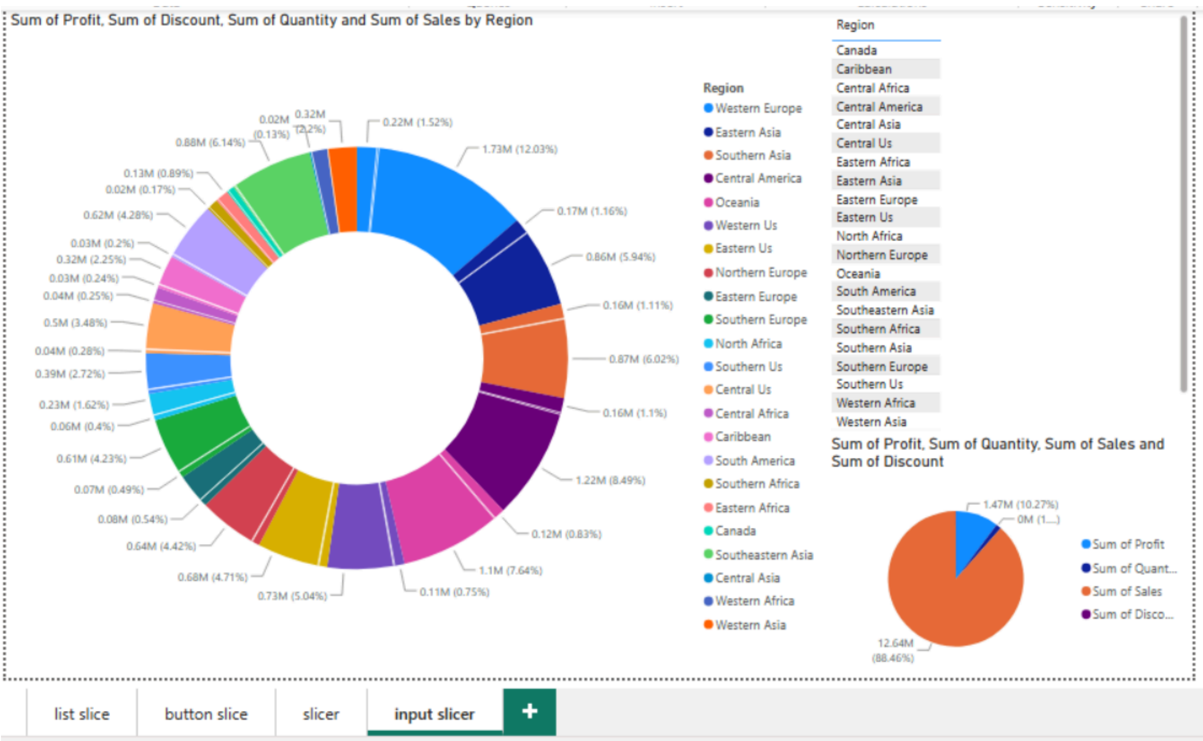


Use the created Dimension Tables to filter the Fact Table and verify that the . relationships work correctly. . Check that changes in dimensions such as Customer, Product, Region, and Date correctly affect the corresponding Fact Table data and dashboard visualizations.

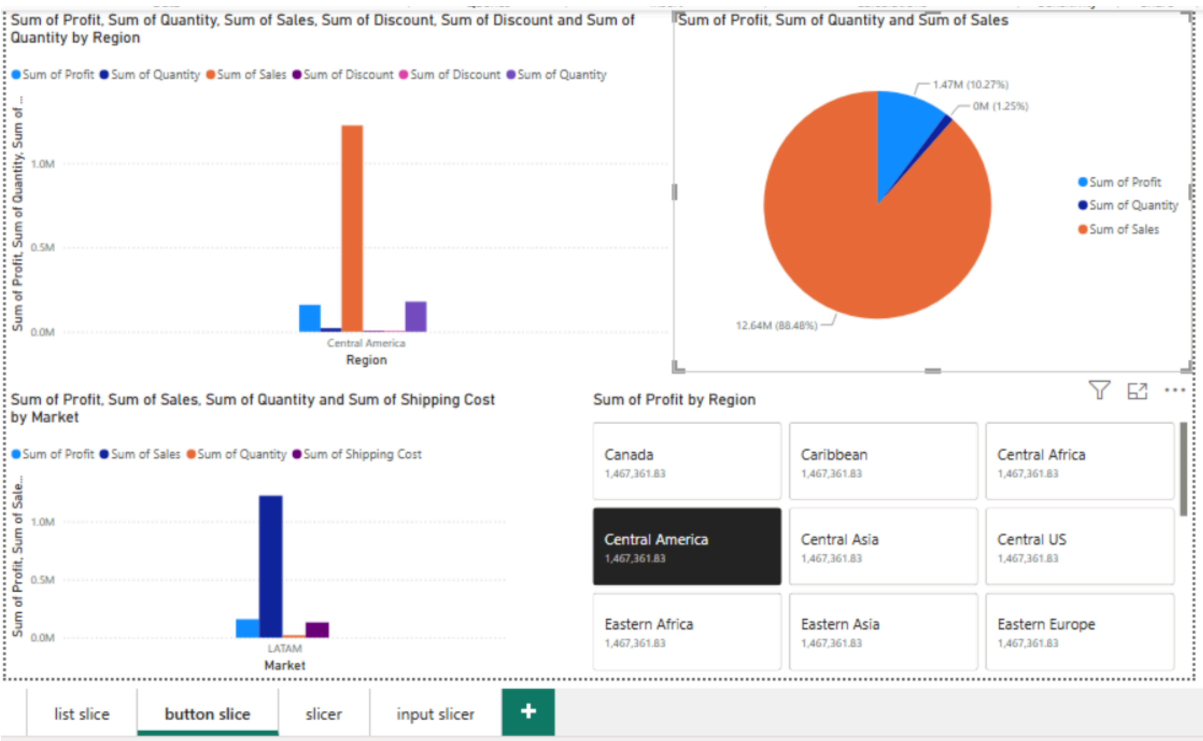
Part G: Slice Operation

Step 22: Slice Operation

Step 23: Input Slice

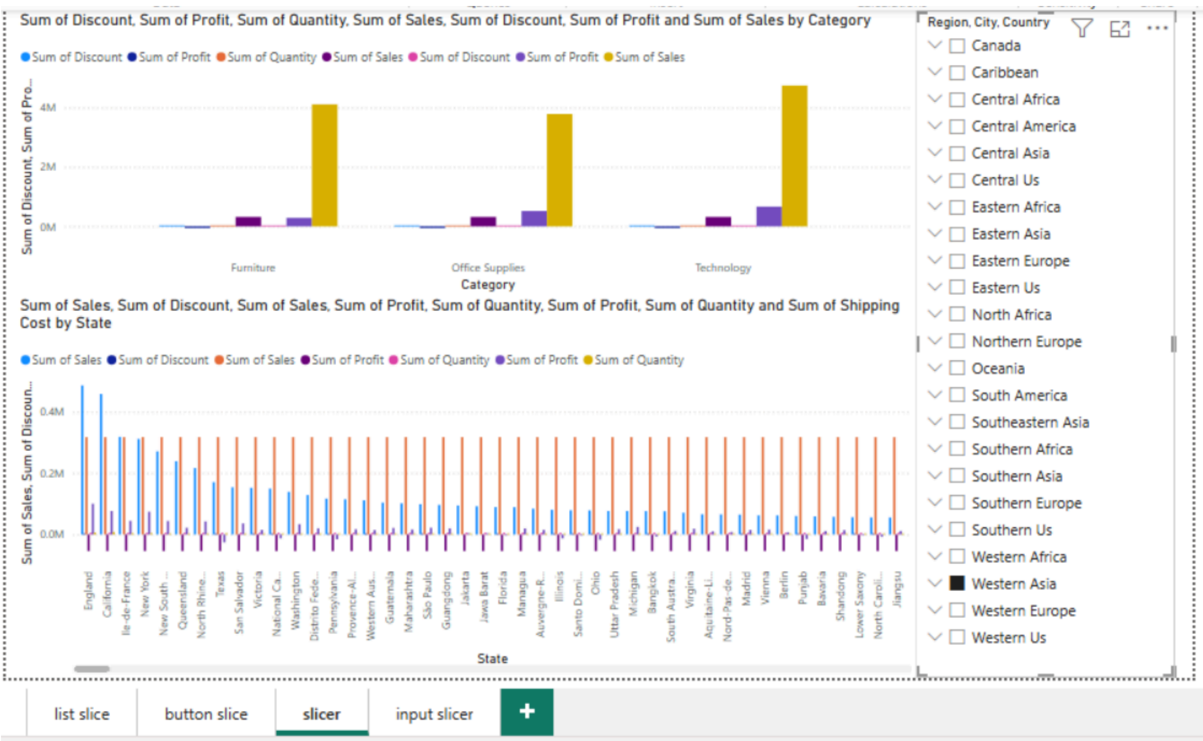


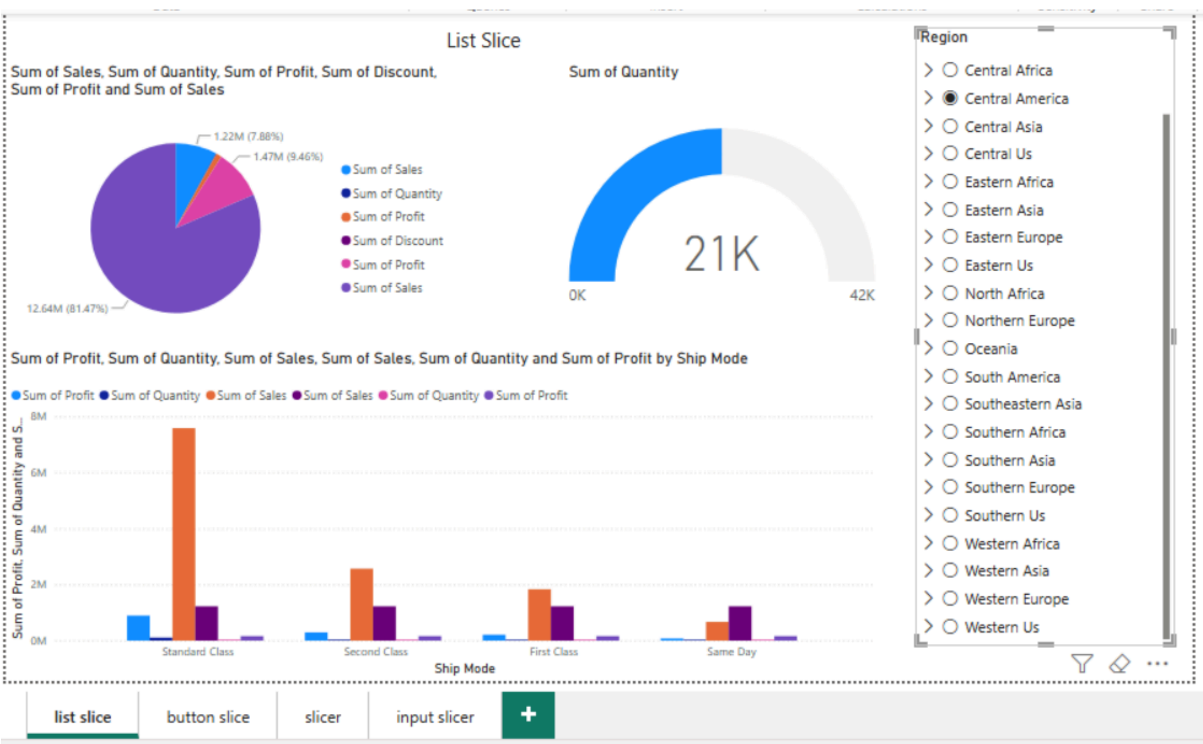
Step 24: Button Slice



Part H: Dice Operation
Step 26: Apply multiple filters simultaneously.
Example:
• Region = West
• Category = Technology
• Year = 2022
Step 27: Analyze filtered business metrics.
Step 28: Record observations and capture screenshots.

Part I: Drill-down Operation
Step 29: Create a hierarchy using Region → State → City.
Step 30: Create a bar chart using the hierarchy.
Step 31: Use Drill-down mode to navigate from Region level to City level.
Step 32: Observe detailed sales patterns.
Step 33: Capture screenshots.
Part J: Drill-up Operation
Step 34: Navigate back from City level to State level and then Region level.
Step 35: Compare aggregated values at each level.
Step 36: Record observations.
Part K: OLAP Analysis
Step 37: Compare results obtained from Slice, Dice, Drill-down, and Drill-up operations.
Step 38: Create a summary table documenting observations.
Step 39: Explain how each operation supports business decision-making.





RESULT:

OLAP operations were successfully performed on the Global Superstore dataset using Microsoft Power BI. Slice, Dice, Drill-down, and Drill-up operations were applied to analyze sales and profit across different dimensions, providing useful insights for business analysis.

LEARNING OUTCOMES:

- Understood the concept and purpose of OLAP in Business Intelligence.
- Learned how multidimensional data can be analyzed from different perspectives.

- Gained practical experience with Slice and Dice operations.
- Learned how to navigate between summarized and detailed information using Drill-down and Drill-up.
- Learned how to use Power BI filters and hierarchies for analytical exploration.
- Understood how sales and profit can be compared across different dimensions.
- Gained experience in creating interactive OLAP-based reports.
- Learned how OLAP analysis can support business decision-making.